

End-to-End Data Analytics Project

Build a complete, portfolio-ready project using Python, SQL, and Power BI to analyze retail customer behavior and drive business decisions.



Project Overview

Real-World Retail Analytics

This comprehensive analytics project analyzes **3,900 customer purchases** across multiple product categories to uncover actionable insights about spending patterns, customer segments, and subscription behavior.

Unlike basic tutorials, this project mirrors the **complete workflow** used by professional data analysts in real organizations—from raw data to executive presentations.

Key Focus Areas

- Customer spending patterns
- Product category performance
- Subscription behavior analysis
- Strategic business recommendations

Why This Project Stands Out



Company-Level Workflow

Experience the complete analytics lifecycle exactly as practiced in real data teams—not just isolated exercises.



Full Tech Stack

Master Python for data processing, SQL for analysis, and Power BI for visualization in one cohesive project.



Portfolio Ready

GitHub documentation, presentation deck, and professional report—everything recruiters want to see.



Business Impact

Learn to translate data insights into strategic recommendations that drive real business decisions.

Complete Project Workflow



Problem Definition

Define clear business objectives based on retail company needs



Data Processing

Clean and transform raw data using Python pandas



SQL Analysis

Run advanced queries to answer business questions



Visualization

Build interactive Power BI dashboards



Communication

Create executive presentations and documentation

Phase 1: Business Problem & Data Preparation

Define the Business Challenge

Start by identifying the **real business problem**: understanding customer behavior to optimize product offerings, improve targeting, and increase revenue.

This phase establishes the foundation for meaningful analysis—exactly how data projects begin in professional settings.

Python Data Processing

Use pandas to load, clean, and transform the retail dataset. Handle missing values, standardize formats, and prepare data for analysis—critical skills for any data analyst role.



Phase 2: SQL Analysis & Insights

Customer Segmentation

Identify high-value customers and purchasing patterns using advanced SQL queries including joins, aggregations, and window functions.

Product Performance

Analyze which product categories drive the most revenue and understand cross-category purchasing behavior.

Subscription Trends

Examine subscription adoption rates, retention patterns, and their impact on customer lifetime value.

Business Metrics

Calculate key performance indicators like average order value, purchase frequency, and customer retention rates.



Phase 3: Power BI Dashboard

Interactive Visualizations

Transform SQL insights into a **beautiful, interactive dashboard** that tells the data story visually.

- Customer demographics and behavior
- Product category performance
- Time-based trend analysis
- Revenue and profitability metrics
- Subscription conversion funnels

Learn to design dashboards that executives can use to make informed decisions—a must-have skill for data analysts.



Phase 4: Documentation & Presentation

Project Report

Create comprehensive documentation covering methodology, findings, and recommendations—exactly what real data teams require.

Executive Presentation

Build a compelling presentation deck that communicates insights clearly to non-technical stakeholders.

GitHub Portfolio

Publish your complete project on GitHub with clear README, code comments, and professional structure.

Skills You'll Demonstrate



100%

End-to-End Workflow

Complete project lifecycle from problem to presentation



3

Core Tools

Python, SQL, and Power BI mastery in one project



3900

Real Data

Transactions analyzed across multiple categories

This project proves you can handle the full scope of data analyst responsibilities—from technical implementation to business communication. Perfect for job applications and portfolio building.

Ready to Build Your Portfolio?



What You'll Achieve

- A complete, **professional-grade project** for your resume
- Hands-on experience with industry-standard tools and workflows
- Portfolio documentation that impresses recruiters and hiring managers
- Confidence to discuss real-world analytics in job interviews

This isn't just another tutorial—it's your launchpad into a data analyst career. Start building today and stand out from other candidates with a project that demonstrates real-world capability.